Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions. The data was cleaned in Python (pandas), analyzed with SQL in MySQL, and presented in an interactive Power BI dashboard.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Loaded the cleaned data into the MySQL database (`analytics`, table `shopping_trends`) for SQL analysis.

4. Data Analysis using SQL (MySQL)

We performed structured analysis in MySQL Workbench to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
1	Female	75191
2	Male	157890

Male customers generate about 68% of revenue (about 2.1x female customers).

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Total rows: 839 (first 10 shown). About half of all discount users (839 of 1,677) still spent at or above the $59.76 average, so discounts do not always mean smaller baskets.

3. **Top 5 Products by Review Rating** – Found the five products with the highest average rating.

	item_purchased	Average Product Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

The top five are only 0.07 points apart, so satisfaction is consistent across products.

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_purchase_amount	num_orders
1	Standard	58.46	654
2	Express	60.48	646

Express customers spend about $2 (3.5%) more per order, with almost the same number of orders.

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Average spend is almost identical, so a subscription does not raise spend per order. Only 27% of customers subscribe.

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Almost half of these products are bought with a discount - above the 43% overall discount rate.

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers	avg_spend	pct_with_discount	avg_rating
1	New	83	58.47	37.3	3.84
2	Returning	701	60.93	40.8	3.77
3	Loyal	3116	59.54	43.6	3.74

80% of customers are Loyal. Discount use rises and ratings fall as customers stay longer.

8. **Repeat Buyers and Subscription** – Checked whether customers with more than 5 previous purchases are more likely to subscribe.

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

Only 27.6% of repeat buyers subscribe - about the same as the overall 27%.

9. **Revenue by Age** – Measured the revenue contribution of each age (top 5 shown).

	age	total_revenue
1	49	5552
2	69	5484
3	25	5372
4	41	5282
5	54	5282

Revenue is spread across all ages; the top age gives only about 2.4% of total revenue.

5. Dashboard using Power BI

We built an interactive Customer Behavior Dashboard in Power BI:

- **KPI Cards:** Number of customers, average purchase amount and average review rating.
- **Subscription Donut Chart:** Share of subscribed vs. non-subscribed customers.
- **Category Charts:** Revenue by category and sales by category.
- **Age Charts:** Revenue by age group.
- **Slicers:** Subscription status, gender, category and shipping type to filter every visual.
- **Design:** Rounded cards, a navy and blue theme and a title banner.



Customer Behavior Dashboard (Power BI)

Dashboard results:

- 3,900 customers, an average purchase of $59.76 and an average review rating of 3.75.
- 73% of customers are not subscribed and 27% are subscribed.
- Clothing is the top category for both revenue and number of purchases, followed by Accessories, Footwear and Outerwear (category values are approximate, read from the charts).
- Revenue is spread evenly across ages 18 to 70, with no single age group dominating.
- The dashboard numbers match the SQL and Python results.

6. Key Insights

1. **Gender:** Male customers generate about 68% of total revenue ($157,890 vs. $75,191).
2. **Discounts:** About half of discount users (839 customers) still spent above the average purchase.
3. **Top-rated products:** Gloves, Sandals, Boots, Hat and Skirt, rated 3.79 to 3.86.
4. **Shipping:** Express customers spend slightly more ($60.48 vs. $58.46).
5. **Subscriptions:** Only 27% subscribe, and subscribers spend the same as non-subscribers ($59.49 vs. $59.87).
6. **Discount-dependent products:** Hat, Sneakers, Coat, Sweater and Pants are bought with a discount 47% to 50% of the time.
7. **Segments:** 80% of customers are Loyal; they use discounts the most (43.6%) and rate lowest (3.74).
8. **Repeat buyers:** Customers with many purchases are not more likely to subscribe.
9. **Age and category:** Revenue is spread evenly across ages; Clothing is the top category.

7. Recommendations

- **Boost subscriptions:** Improve the subscription offer and target the 2,518 repeat buyers who have not subscribed.
- **Review discount dependence:** Test loyalty rewards that do not rely on price cuts, especially for Loyal customers and discount-dependent products.
- **Reach female customers:** They bring in about a third of revenue; test campaigns and product ranges for them.
- **Promote Express shipping** to higher-value baskets.
- **Protect satisfaction among loyal customers:** Ratings fall with tenure (3.84 to 3.74).